

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

A PROJECT REPORT ON PHISHING EMAIL DETECTION, THREAT
ANALYSIS AND SECURITY AWARENESS

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INTRODUCTION

Phishing is one of the most common cyberattacks used by attackers to steal sensitive information such as usernames, passwords, banking details, and personal data through fraudulent emails and fake websites. These phishing emails are designed to appear legitimate and often use social engineering techniques to trick users into clicking malicious links or downloading harmful attachments.

The main objective of this project is to analyze phishing email samples, identify phishing indicators, classify suspicious emails, and create cybersecurity awareness among users. This project focuses on email header analysis, malicious link detection, sender verification, and phishing prevention techniques using various cybersecurity tools and public phishing datasets.

The project also highlights the importance of security awareness and safe email practices to reduce the risk of phishing attacks in organizations and daily digital communication.

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OBJECTIVES

- To collect and analyze phishing email samples using cybersecurity analysis techniques.
- To identify phishing indicators such as spoofed sender addresses, fake domains, malicious links, suspicious attachments, and social engineering tactics.
- To perform email header analysis and investigate suspicious email activities.
- To classify emails as Safe, Suspicious, or Phishing based on the identified threats and risk level.
- To create cybersecurity awareness by explaining phishing attacks and prevention methods in simple and professional language.
- To provide security recommendations and safe email practices for users and organizations.
- To document the complete phishing analysis process and submit the findings through a professional report and public GitHub repository.

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TOOLS USED

Tool	Purpose
VirusTotal	URL reputation analysis
MXToolbox	Email header analysis
Google Admin Toolbox	Header tracing
GitHub	Project hosting
Browser Developer Tools	Link inspection

METHODOLOGY

Step 1: Collection of Email Samples

Phishing and legitimate email samples were collected from public phishing datasets and cybersecurity resources such as Kaggle, PhishTank, and phishing email repositories.

Step 2: Email Examination

The collected email samples were manually examined to identify suspicious characteristics including spoofed sender addresses, fake domains, urgent language, suspicious attachments, and malicious links.

Step 3: Header Analysis

Email headers were analyzed using email header analysis tools to inspect sender details, routing paths, SPF/DKIM authentication status, and possible spoofing attempts.

Step 4: Link and Attachment Analysis

Suspicious links and attachments found in the emails were analyzed using online threat intelligence and reputation-checking tools to determine whether they were malicious or safe.

Step 5: Identification of Phishing Indicators

Different phishing indicators such as fake branding, social engineering techniques, credential harvesting attempts, and suspicious formatting were identified and documented.

Step 6: Email Classification

Each email sample was classified into one of the following categories based on the analysis results:

- Safe
- Suspicious
- Phishing

Step 7: Risk Assessment

The identified phishing emails were evaluated based on their potential impact on users and organizations. The analysis focused on risks such as credential theft, unauthorized account access, malware infection, financial fraud, identity theft, and data breaches.

Step 8: Documentation and Reporting

All findings, screenshots, analysis results, phishing indicators, and prevention guidelines were documented in a professional phishing detection and awareness report.

Step 9: GitHub Repository Submission

The final report, screenshots, email samples, and supporting documents were organized and uploaded to a public GitHub repository for project submission and documentation purposes.

Email Analysis

Sample-1-Netflix Payment Scam Phishing Email

Email Overview

From	Netflix <replynet30159@nita.tc>
To	jose@monkey.org
Subject	Please update your payment method!
Reply-To	replynet30159@nita.tc
Classification	PHISHING
Risk Level	HIGH

Email Body (Reconstructed Sample)

Dear Customer,

We could not process your Netflix payment. Your account may be suspended unless you update your payment information immediately.

Update Payment Method:
<https://eliteakademidiyarbakir.com/update-your-account/>

Thank you,
Netflix Billing Team

Phishing Indicators Identified

Indicator	Classification	Risk Level	Recommendation
Fake Netflix sender address	Spoofed Identity	High	Verify sender domain before trusting emails
Suspicious external URL	Malicious Link	High	Never click links from unknown domains
Urgent payment warning	Social Engineering	High	Attackers use fear and urgency to pressure victims
Generic greeting (“Dear Customer”)	Mass Phishing Campaign	High	Legitimate companies usually personalize emails
Domain mismatch with Netflix	Brand Impersonation	High	Official Netflix emails come from netflix.com domains
Request to update billing details	Credential Theft Attempt	High	Never enter credentials through emailed links

Sample-2- Legitimate Netflix Billing Notification

Email Overview

Field	Details
From	Netflix <support@netflix.com>
To	jose@monkey.org
Subject	Your Netflix Payment Receipt
Classification	SAFE EMAIL
Risk Level	LOW

Email Body (Reconstructed Sample)

Dear Jose,

Your monthly Netflix subscription payment has been successfully processed.

You can review your billing details and account information anytime through the official Netflix website.

<https://www.netflix.com/account>

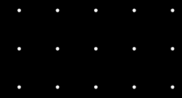
Thank you for being a valued Netflix member.

Netflix Billing Team

Safe Email Indicators Identified

Indicator	Classification	Risk Level	Observation
Official sender domain	Legitimate Domain	Low	The email uses the official netflix.com domain
Professional communication style	Legitimate Communication	Low	The message is clear, professional, and properly structured
No urgent threats or panic language	Safe Behavior	Low	The email does not pressure the user into immediate action
Valid official hyperlink	Trusted URL	Low	The link redirects to the official Netflix website
Personalized customer communication	Legitimate Service Notification	Low	The recipient name and billing details are communicated professionally
No credential harvesting attempt	Safe Authentication Practice	Low	The email does not request passwords or sensitive information through suspicious links

RISK ASSESSMENT



Phishing emails can cause serious security risks such as credential theft, malware infection, financial fraud, and unauthorized account access. During the analysis process, suspicious sender addresses, fake domains, malicious links, and phishing techniques were identified in several email samples.

Based on the identified phishing indicators, the emails were classified as Low, Medium, or High risk. This assessment highlights the importance of cybersecurity awareness and safe email practices to prevent phishing attacks.

PREVENTION GUIDELINES

- Verify sender email addresses carefully.
- Avoid clicking suspicious links or attachments.
- Enable multi-factor authentication (MFA).
- Use updated antivirus and email security tools.
- Report suspicious emails immediately.

CONCLUSION

This project successfully analyzed phishing email samples and identified various phishing indicators such as spoofed sender addresses, fake domains, malicious links, and suspicious attachments. The emails were classified based on their risk level as Safe, Suspicious, or Phishing.

The project highlights the importance of cybersecurity awareness, email verification, and safe browsing practices to reduce the risk of phishing attacks and improve email security.

REFERENCES

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